

Maheshwar Rao Bandi

Generative AI Engineer | Machine Learning Engineer | Data Scientist

Kansas, USA | +1 (913) 353-5764 | maheshwarraobandi@gmail.com | [LinkedIn](#)

PROFESSIONAL SUMMARY

AI/ML Engineer and Data Scientist with 5+ years across applied ML, deep learning, and analytics — including 2+ years shipping production Generative AI systems in financial services, logistics, pharmaceuticals, and insurance. Specialized in LLM-powered applications (RAG, agentic workflows, fine-tuning with LoRA/QLoRA), LLM evaluation and observability (Ragas, LangSmith), classical and deep ML (XGBoost, Transformers, CNNs/RNNs, time-series), and multi-cloud MLOps on AWS, Azure, and GCP. Track record of translating ambiguous business problems into measurable outcomes — 45% support deflection at JPMC, \$1.8M annual logistics savings, \$50M+ pricing decisions informed by models, and 30–70% efficiency gains across automation, forecasting, and CX.

CORE TECHNICAL COMPETENCIES

Generative AI & LLMs: GPT-4 / 4o, Claude 3, Llama 3, Mistral, BERT, T5, RAG, Agentic Workflows, LoRA / QLoRA / PEFT, RLHF, Prompt Engineering, Few-/Zero-shot, Multimodal AI, Stable Diffusion, DALL·E

LLM Frameworks & Tooling: LangChain, LangGraph, LangSmith, LlamaIndex, Hugging Face Transformers, OpenAI API, Anthropic API, Cohere, vLLM, Gradio, Streamlit, FastAPI

Vector Databases & Retrieval: Pinecone, FAISS, Weaviate, ChromaDB, Milvus, Hybrid search, Reranking, Semantic chunking

ML / Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, CNNs, RNNs / LSTMs, Transformers, GANs, VAEs, Autoencoders, Diffusion, Regression, SVM, KNN, Random Forest, Gradient Boosting, K-Means, Collaborative Filtering

NLP: Transformers, NLTK, spaCy, Gensim, Sentiment analysis, NER, Topic modeling, Text classification

Data Engineering & Big Data: Apache Spark (PySpark, Spark SQL, MLlib), Hadoop (HDFS, Hive, HBase), Apache Airflow, Kafka, Flume

Cloud & MLOps: AWS (SageMaker, Bedrock, Lambda, EC2, S3, EKS), Azure (OpenAI, ML Studio, Cognitive Services), GCP (Vertex AI, BigQuery), Docker, Kubernetes, MLflow, DVC, Weights & Biases, GitHub Actions, Jenkins

Programming & Databases: Python, SQL, R, Java, C, Bash, PostgreSQL, MySQL, Oracle, SQL Server, MongoDB, Neo4j

Visualization & BI: Tableau, Power BI, Plotly, Matplotlib, Seaborn, ggplot2

PROFESSIONAL EXPERIENCE

JP Morgan Chase | Topeka, KS, USA

Mar 2025 — Present

Generative AI Engineer

- Architected and deployed an enterprise-scale GenAI customer-support assistant on GPT-4 with hybrid retrieval (Pinecone semantic search + FAISS reranking) over 500K+ documents — sub-200ms p95 retrieval latency, 45% lower average ticket resolution time, and 30% L1 ticket deflection.
- Fine-tuned Llama 3 and BERT with LoRA / QLoRA on 2M tokens of proprietary financial documentation, lifting domain QA accuracy from 68% to 91% and cutting inference cost 40% via 4-bit quantization.
- Built an LLM observability stack (LangSmith + MLflow + Grafana) tracking token usage, hallucination rate (Ragas), and prompt drift across 12+ production agents — improving release safety and cutting bad-output incidents in early production.
- Productionized generative-image pipelines (Stable Diffusion, GANs) for marketing creatives and packaging mock-ups, replacing \$120K/yr in external design spend.
- Deployed FastAPI microservices on AWS EKS with autoscaling and blue/green CI/CD via GitHub Actions and Docker, sustaining 5K+ requests/min at 99.9% uptime; integrated Azure Cognitive Services for OCR, content moderation, and image tagging into KYC workflows; delivered SHAP-based explainability dashboards for compliance and risk stakeholders.

Tech Stack: OpenAI API, Anthropic Claude, LangChain, LangSmith, FAISS, Pinecone, Hugging Face, Stable Diffusion, GANs, Azure Cognitive Services, FastAPI, Docker, Kubernetes, AWS EKS / SageMaker / Bedrock, MLflow, Weights & Biases, Streamlit, Grafana, Tableau

GXO Logistics | Olathe, KS, USA

Aug 2024 — Feb 2025

Machine Learning Engineer

- Designed and shipped demand-forecasting and route-optimization ML models (XGBoost, Random Forest, LSTM) on GCP Vertex AI, lifting forecast accuracy by 22% and saving \$1.8M annually in logistics overhead.
- Built an NLP-powered shipment-classification engine (BERT + spaCy) processing 1M+ daily logistics records, replacing manual triage and reducing routing errors by 35%.
- Engineered scalable PySpark + GCP BigQuery + Apache Airflow data pipelines processing 10TB+/day at 99.5% SLA adherence.

- Established the MLOps backbone with MLflow, DVC, GitHub Actions, and Vertex AI Model Registry — cutting model deployment cycle from 3 weeks to 2 days.
- Containerized ML services with Docker and Kubernetes (GKE), exposed REST APIs (FastAPI) consumed by warehouse-management systems and led sprint planning for a 3-engineer pod against a prioritized AI roadmap.

Tech Stack: Python, TensorFlow, Keras, XGBoost, FB Prophet, BERT, spaCy, GCP Vertex AI, BigQuery, Apache Airflow, PySpark, AWS SageMaker, Docker, Kubernetes, MLflow, DVC, GitHub Actions, Tableau

Ajanta Pharma Limited | Mumbai, India

May 2021 — Jul 2023

Data Scientist

- Developed deep-learning models (CNNs, RNNs, Transformers) for pharmaceutical image classification and adverse-event text mining, achieving 94% F1 and accelerating regulatory review cycles by 28%.
- Built end-to-end demand-forecasting and price-elasticity models (Random Forest, Gradient Boosting, ARIMA, Linear / Logistic Regression) that informed \$50M+ of annual inventory and pricing decisions.
- Designed scalable data pipelines on Hadoop (HDFS, Hive, HBase) and Spark (PySpark, MLlib), unifying PostgreSQL, MySQL, Oracle, MongoDB, and Neo4j sources into a 5TB+ analytical lake.
- Productionized 15+ ML models via Docker, Kubernetes, MLflow, DVC, and Jenkins on AWS SageMaker, GCP Vertex AI, and Azure ML, with automated drift detection and retraining.
- Delivered customer-segmentation and recommendation systems (K-Means, Hierarchical Clustering, Collaborative Filtering) that boosted targeted-campaign conversion by 18%; built executive Tableau and Plotly dashboards consumed in C-suite monthly business reviews.

Tech Stack: Python, R, SQL, TensorFlow, PyTorch, Hadoop, Spark, Kafka, Flume, Tableau, Docker, Kubernetes, MLflow, DVC, Jenkins, AWS SageMaker, GCP Vertex AI, Azure ML

ICICI Prudential Life Insurance | Mumbai, India

Mar 2020 — Apr 2021

Data Scientist

- Delivered 13+ analytics projects across portfolio modeling, targeted marketing, KPI forecasting, and enterprise data products within a 20-person cross-functional program.
- Drove digital strategy that lifted digital transactions by 15% and improved conversion by 10%; re-architected the reporting stack with self-service BI, cutting decision-making turnaround by 70%.
- Deployed NLP pipelines (NLTK, spaCy, Transformers, RNN / LSTM) for sentiment analysis, topic modeling, and NER on customer-support and policy data.
- Productionized ML models on Azure ML, AWS SageMaker, and GCP Vertex AI using Docker, Kubernetes, Jenkins, and GitHub Actions — enabling automated, scalable AI services.

Tech Stack: R, Python, SQL Server, Tableau, Azure ML, AWS SageMaker, GCP Vertex AI, Docker, Kubernetes, Jenkins, GitHub Actions

KEY PROJECTS

Multi-Agent Financial Research Assistant

- Built an agentic GenAI system (LangGraph + GPT-4 + Claude) with tool-using agents, structured-output parsing, self-reflection loops, and Pinecone-backed long-term memory orchestrating SEC-filings retrieval, statement analysis, and competitor benchmarking — cutting research time from ~4 hours to ~12 minutes.

Tech Stack: LangGraph, OpenAI GPT-4, Anthropic Claude, Pinecone, FastAPI, Streamlit, Docker

Domain-Tuned Medical QA Chatbot

- Fine-tuned Llama 3 8B with QLoRA on a curated PubMed + clinical-guideline corpus and deployed a RAG-augmented chat UI with hallucination guardrails, PII redaction, and citation grounding — improved factuality (Ragas) from 0.71 -> 0.89.

Tech Stack: Llama 3, Hugging Face PEFT / Transformers, ChromaDB, LangChain, Gradio, AWS EC2

EDUCATION

Master of Science in Data Science — University of Missouri, Kansas City, USA

GPA: 3.6 / 4.0 | May 2025